Products of Cosines: How the Entangling Layers of Quantum Classifier Heads Shape the CNN Representations Trained Through Them

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Abstract

When a convolutional network is trained end to end through a variational quantum circuit (VQC) classifier head, the head shapes the features the network learns, and we measure this with one common linear probe on the frozen bottleneck of every trained network. On CIFAR-10 with 4,000 training images, an 8-qubit VQC head with CNOT rings lowers probe accuracy by 4.7 points relative to a linear head (5 seeds, every seed lower; a two-layer MLP probe finds the same gap), and removing the rings recovers 3.8 of those points. An exact identity organizes the pattern: with every variational rotation at zero, each readout is a product of cosines of the encoded inputs over a set fixed by the CNOT layers. Probe accuracy falls as the mean order of these products rises, from 73.6 at order 1 to 69.8 at the default order 2.5 and 60.1 at order 5.25; classical heads trained on the same products follow the same trend, the circuit ordering holds at twice the training budget and on MNIST, designed classical supports favor lower order at fixed mean overlap between readouts, and an entangling circuit with product order 1 recovers 3.1 points. On an ImageNet-pretrained ResNet-18 the same ordering holds, and the default head's cost grows to 22 points with 1,000 training images while the ring-free head stays within 1.6 points of a linear head. Without CNOTs the head is exactly a classical per-coordinate sinusoidal model, and under matched per-head tuning a linear head has the highest mean test accuracy among linear, small-MLP and default VQC heads in every setting we test.

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1. Introduction

A common near-term design for quantum image classification feeds a convolutional network into a small variational quantum circuit (VQC) that acts as the classifier head (Mitarai et al., 2018; Havlíček et al., 2019; Benedetti et al., 2019; Cerezo et al., 2021). The network is trained *through* the circuit, so the head does more than classify: its gradients shape the features the backbone learns. A comparison between a quantum head and a classical one therefore compares two things at once, the heads and the representations they induce. Benchmarks of quantum classifiers read fixed inputs (Bowles et al., 2024; Fulton et al., 2026; Slattery et al., 2022), and the controls that isolate a circuit inside a hybrid model score output accuracy (Kölle et al., 2024; Elhag et al., 2026). Neither tells us what the quantum head does to the representation trained through it, or which part of the circuit is responsible.

We answer both questions for a CNN with an 8-dimensional bottleneck and an 8-qubit head built from angle encoding $R_y(\pi \tanh z_j)$, four layers of trainable rotations each followed by a ring of CNOTs, and a linear readout of $\langle Z_k \rangle$. After joint training we freeze each backbone and fit the same linear probe to its bottleneck, so that probe accuracy compares the representations directly. The default VQC head lowers this probe accuracy, and the size of the drop follows the circuit's CNOT structure. An exact identity explains why structure matters: when every trainable rotation is zero, which is near where training starts, each readout is $\langle Z_k \rangle = \prod_{j \in S_k} \cos(\pi \tanh z_j)$, a product of cosines over a set S_k of inputs fixed by the CNOT layers alone (Figure 1a). The mean size of these sets, the *product order*, is 1 without CNOTs, 2.5 for the default circuit and 5.25 when a single ring follows the last layer.

Our contributions are:

1. **A tuning audit** (Section 4). Under a revised protocol (longer training, validation-based selection, each head's learning rate tuned), the VQC's apparent advantage over a near-parameter-matched MLP on MNIST shrinks from up to 18 points to a largest observed mean of 3.3. On CIFAR-10 the VQC still trails the MLP by 3.1–9.2 points, and among the linear, MLP and default

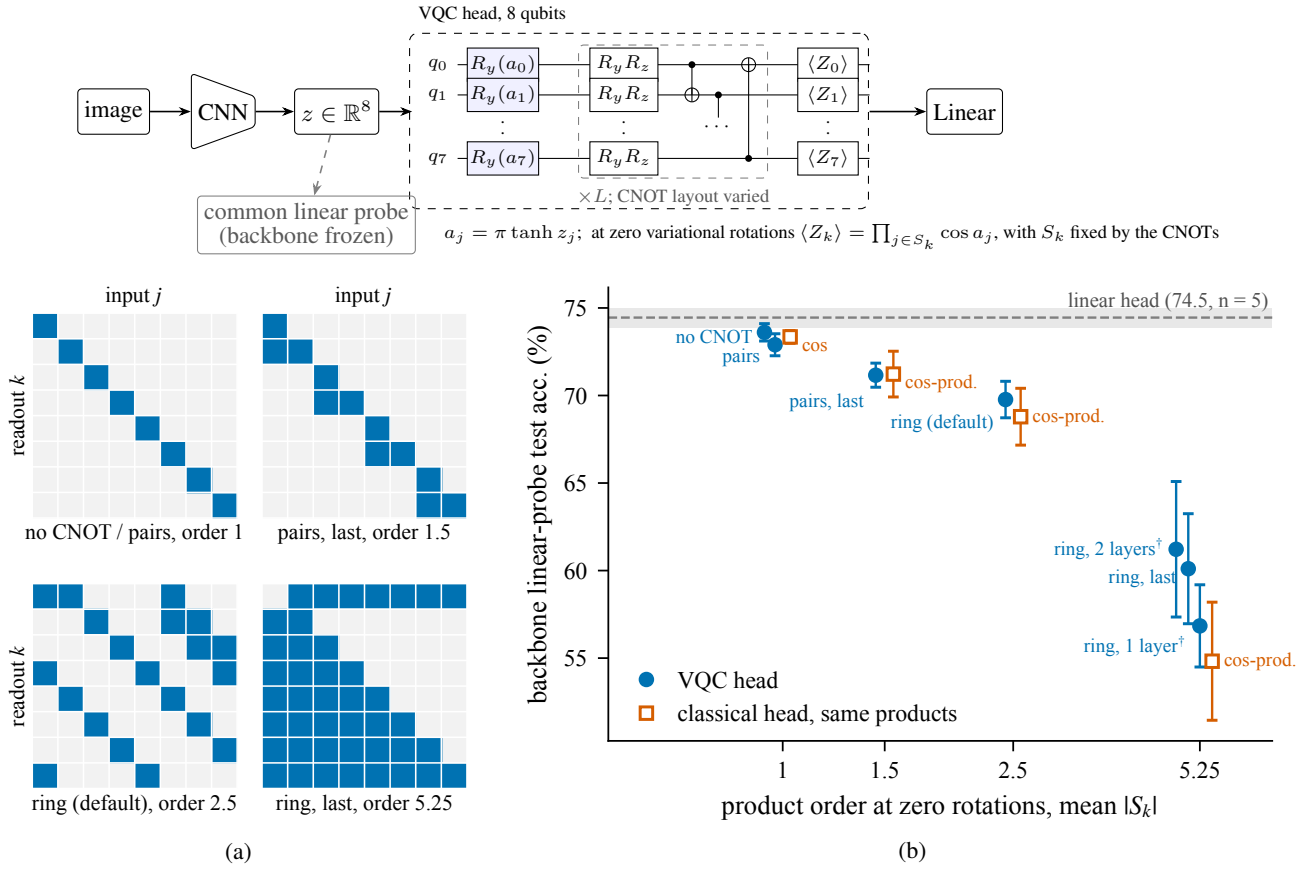


Figure 1. **(a)** A CNN feeds an 8-dimensional bottleneck z into an 8-qubit VQC head (default: four $R_y R_z$ layers, each followed by a CNOT ring $(q, q{+}1 \bmod 8)$; variants place CNOTs after the last layer only ("last"), use $L \in \{1, 2\}$ layers, or replace the ring by disjoint pairs $(0,1), (2,3), (4,5), (6,7)$). With every variational rotation at zero, each readout is exactly $\langle Z_k \rangle = \prod_{j \in S_k} \cos(\pi \tanh z_j)$; the grids mark $j \in S_k$ in blue (rows: readouts k, columns: inputs j). Pairs after every layer cancel at zero rotations, so their supports equal the no-CNOT ones. Product order is $\frac{1}{8} \sum_k |S_k|$. The backbone is evaluated by one common linear probe on the frozen bottleneck. **(b)** CIFAR-10 with 4,000 training images: probe test accuracy against product order (log axis) for VQC heads (filled) and classical heads trained end to end on the zero-rotation products of a circuit (open; at order 5.25 these are the supports of "ring, last" and "ring, 1 layer"; "ring, 2 layers" has different supports of the same mean order). Mean ± SD over 5 seeds (†: 3); head learning rate selected on mean validation accuracy over seeds 0–2, seeds 3–4 run at that choice. Dashed line and band: linear head. Points at one order are offset horizontally. Order is a zero-rotation descriptor that co-varies with support overlap; the plot shows an association, and the classical heads reproduce the declining trend, closely for the default ring but 5 points lower for "ring, last".

VQC heads the linear head has the highest mean test accuracy in all six dataset–size settings.

2. **A representation cost tied to the CNOT layers** (Sections 5.1 and 5.2). On CIFAR-10 with 4,000 training images the default VQC head lowers bottleneck probe accuracy by 4.7 ± 0.9 points relative to a linear head (5 seeds), and a two-layer MLP probe finds the same gap, so the loss is not just one of linear readability. Removing the CNOT rings recovers 3.8 ± 1.2 points. On a pretrained ResNet-18 the default head costs 7.2 ± 2.3 points at 4,000 images and 22.3 at 1,000, while the ring-free head matches the linear one (Section 5.6). Five controls—backbone learning rate, frozen rotations, pre-encoding BatchNorm and two alternative encodings—leave a substantial gap (Section 5.5).

3. **A structural account** (Sections 5.3 and 5.4). Probe accuracy falls with product order across eleven heads (Figure 1b). Classical heads trained on a circuit's zero-rotation products, with no circuit, follow the same trend; on the small CNN the default circuit's products match the VQC (-1.0 ± 2.2), and on the pretrained backbone they reproduce the ordering but only part of the cost. Classical heads with designed supports favor lower order at fixed mean overlap between readouts and full rank, though order alone does not set the cost, and the ordering replicates on MNIST under predictions fixed before the runs. A circuit that keeps its CNOTs but has product order 1 recovers 3.1 ± 1.2 points over the default. Without CNOTs the head is exactly a classical per-coordinate sinusoidal model, and classical

heads of that form come within 0.3–0.9 probe points of it.

The practical upshot is concrete: in this design, the entangling layout of a quantum head is a choice about the backbone's representation, and product order, computable from the CNOT pattern before training, orders the mean probe accuracy of every circuit we test. Section 2 positions the work, Section 3 gives the model, protocol and identity, Sections 4 and 5 report the results, and Section 6 discusses a mechanism hypothesis and the limits of the evidence.

2. Related Work

Benchmarking quantum classifiers and the role of entanglement. Large benchmarks find that simple classical models generally match or beat variational quantum classifiers, and that removing entanglement often does not hurt (Bowles et al., 2024); quantum fidelity kernels show a similar lack of advantage on classical data (Slattery et al., 2022). Closest to our setting, Fulton et al. (2026) show that classical $\mathrm{SU}(2)$ models match or exceed shallow VQCs on vision benchmarks and that ring entanglement can degrade accuracy on fixed features. Recent hybrid-network studies reach related conclusions under more careful matching of width and budget (Freinberger & Moser, 2026; Innan et al., 2026; Park et al., 2026). Other work trains hybrid models end to end and studies the learned representation: Kashif & Shafique (2026) analyze expressibility and trainability under joint training, and quantum self-supervised learning evaluates frozen hybrid encoders with linear probes (Jaderberg et al., 2021; Edwards et al., 2026). We probe the supervised bottleneck *before* the circuit, the representation the circuit's gradients shape, and relate its accuracy to a quantity, product order, read off the CNOT pattern.

Isolating the circuit inside hybrid models. A growing line of work asks what the quantum part of a hybrid model contributes. Kölle et al. (2024) separate classical and quantum contributions in hybrid architectures, building on sequential training that first fits a classical surrogate and then swaps in the circuit (Altmann et al., 2023). Elhag et al. (2026) delete, freeze or corrupt the circuit in quantum message passing and compare it with exact classical surrogates under the same $\pi\tanh$-style angle encoding. Data-encoding choices in hybrid convolutional models have also been compared directly (Monnet et al., 2024). These controls score predictive accuracy. We add a representation-level measurement, a common probe on the jointly trained bottleneck, and controls that change one structural property of the circuit at a time.

Fourier structure, classical surrogates and head-induced feature distortion. Angle-encoded circuits compute truncated Fourier series of their inputs (Schuld et al., 2021; Pérez-Salinas et al., 2020), which classical models can represent and emulate (Schreiber et al., 2023); sinusoidal features also appear in classical networks as periodic activations and Fourier feature maps (Sitzmann et al., 2020; Tancik et al., 2020). Our zero-rotation identity is an elementary instance of this structure, used as an explanatory variable: it shows which products the CNOT layers create, and their order tracks a measured cost. In classical deep learning, Kumar et al. (2022) show that the head used during fine-tuning distorts pretrained features, and linear probes are a standard tool for reading what an intermediate layer encodes (Alain & Bengio, 2016). We apply the probe to the quantum-head setting, where the head is a fixed-form circuit rather than a learned classical layer.

Position of this work. That entanglement often does not help, and that trigonometric surrogates can stand in for simple circuits, are known. What is new here is (i) a measurement of the bottleneck a quantum classifier head induces under supervised joint training, with a probe shared across quantum and classical heads, and (ii) a structural variable, the order of the zero-rotation cosine products, that orders the representation cost across circuit and classical heads and survives an intervention that keeps entanglement while lowering order.

3. Setup

Model. A CNN backbone maps an image to a bottleneck $\mathbf{z} \in \mathbb{R}^8$ (MNIST: three convolutional blocks, about 94K parameters; CIFAR-10: four blocks, about 0.96M; Section 5.6 also uses an ImageNet-pretrained ResNet-18). Each head is trained jointly with its own copy of the backbone. The *linear* head is a single dense layer (90 parameters); *MLP-8* has one hidden layer of width 8 with ReLU (162). The default *VQC* head encodes $a_j = \pi \tanh z_j$ by $R_y(a_j)$ on qubit j, applies $L = 4$ layers of trainable $R_y R_z$ rotations on every qubit, each followed by a ring of CNOTs $(q, q{+}1 \bmod 8)$ applied in order $q = 0, \ldots, 7$, measures $\langle Z_k \rangle$ on every qubit, and maps the eight expectations to ten logits with a linear layer (154 stored parameters, 146 of which can affect the output). Rotations are initialized in $\mathcal{U}(-0.1, 0.1)$. We simulate the circuit exactly with a batched PyTorch statevector simulator that matches PennyLane (Bergholm et al., 2018) to 10^{-6} in outputs and input gradients and 10^{-5} in rotation gradients (our test suite). Circuit variants change CNOT placement or rotation depth: *no CNOT*; a ring after the *last* layer only; $L \in \{1, 2\}$ layers; and disjoint *pairs* $(0,1), (2,3), (4,5), (6,7)$ after every layer or after the last layer only.

Protocol. Training subsets are class-stratified samples of 250, 500 or 1,000 MNIST images and 1,000, 2,000 or 4,000

CIFAR-10 images; a disjoint stratified 2,000-image validation set selects the checkpoint epoch. Every run uses AdamW (weight decay 10^{-4}), a cosine schedule, batch size 32 and gradient clipping, for 50 epochs on MNIST and 80 on CIFAR-10 with standard augmentation. The backbone learning rate is half the head learning rate. For each dataset, size and head we select the head learning rate with the best mean validation accuracy over seeds 0–2 from a grid of half-decade steps, and report test accuracy once. Headline CIFAR-10/4,000 cells add seeds 3 and 4 at that selected rate. Heads in a cell share seeds, hence training subsets and validation splits, so we report paired differences (mean $\pm$ SD over seeds, in percentage points). Every table states its number of seeds.

Common linear probe. To compare representations, we freeze each selected run's backbone, extract $\mathbf{z}$ on clean images, and fit a fresh linear layer $\mathbb{R}^8 \to \mathbb{R}^{10}$ (AdamW, cosine schedule, probe learning rate and epoch selected on validation). Because the probe is the same for every backbone, differences in its test accuracy compare the bottlenecks that different heads induced in joint training. We call this *probe accuracy*.

Zero-rotation identity. With every trainable rotation set to zero, the variational block reduces to its CNOTs. The encoded state is a product state with $\langle Z_j \rangle = \cos a_j$, and conjugating by $\mathrm{CNOT}(c,t)$ maps $Z_t \mapsto Z_c Z_t$ and leaves Z_c unchanged. Propagating Z_k back through the gates in reverse order, and cancelling repeated factors with $Z^2 = I$, gives a Pauli-Z string on a set S_k of qubits:

$$\langle Z_k \rangle\big|_{\theta=0} = \prod_{j \in S_k} \cos(\pi \tanh z_j), \qquad \bar{s} = \tfrac{1}{8} \textstyle\sum_k |S_k|. \tag{1}$$

We call $\bar{s}$ the *product order*. It is 1 without CNOTs, 2.5 for the default circuit, 5.25 for a ring after the last layer or a single-layer circuit, 1.5 for pairs after the last layer, and 1 for pairs after every layer: the pair layer P satisfies $P^2 = I$, so its four copies cancel at zero rotations, while interleaved nonzero rotations generally prevent the cancellation (Figure 1a; Section A lists the supports). Without CNOTs the identity extends to all rotation angles: each qubit evolves alone, so

$$\langle Z_j \rangle = \alpha_j \cos a_j + \beta_j \sin a_j, \tag{2}$$

with α_j, β_j set by qubit j's trained rotations. With its readout layer, the ring-free head is thus a classical per-coordinate sinusoidal model (we checked the formula against the simulator to 10^{-7}). To test whether the products themselves carry the effect, we train classical heads on them with no circuit: *cos* uses $\cos a_j$ (order 1), *cos+sin* adds $\sin a_j$, and *cos-prod.* uses the products of Equation (1) for a given circuit's supports, each followed by a linear layer and tuned like every other head.

4. Tuning Changes Head Comparisons, but a Linear Head Stays Ahead

Before turning to representations, we check how much a head comparison depends on the training recipe. In an exploratory sweep with one shared recipe (head learning rate 10^{-3} for every head, 25 MNIST epochs; Section E), the VQC beat MLP-8 on MNIST by 17.8, 18.0 and 6.9 points at 250, 500 and 1,000 images and lost to it on CIFAR-10 by 14.1, 13.8 and 8.3 points. Under the protocol of Section 3 (50 MNIST epochs, validation-based selection, per-head learning rates), these differences change in both size and sign (Table 1). On MNIST, VQC $-$ MLP-8 becomes -2.1 ± 9.7, $+3.3 \pm 2.6$ and $+0.8 \pm 1.0$. Part of this change comes from the protocol itself: at the shared rate under the new protocol the advantage is already $+10.8$, $+6.1$ and -0.5 (last column of Table 1), and per-head tuning moves it to the values above. On CIFAR-10, where only the learning rate changes between the two columns of Table 1, the VQC deficit to MLP-8 shrinks from 10.1–13.6 points to 3.1–9.2, and the VQC's own test accuracy rises by 6.0, 6.1 and 10.7 points. A linear head has the highest mean in all six settings and beats the VQC for every seed; its mean lead is 1.7–13.7 points on MNIST and 7.0–15.8 on CIFAR-10. Its selected rate sits at the top of its grid in five settings, so its lead may be understated.

This audit fixes the comparisons used in the rest of the paper: every head is evaluated at its own validated learning rate. It also leaves a question. At 4,000 CIFAR-10 images the VQC trails the linear head by 7.0 ± 1.2 test points (5 seeds: 6.7 ± 1.0). Part of that gap could come from the VQC as a readout and part from the backbone it trains; a common probe measures the gap that remains under one readout.

5. What the VQC Head Does to the Backbone

5.1. The default VQC head lowers bottleneck probe accuracy

On CIFAR-10 with 4,000 images, the common probe reaches 69.8 ± 1.0 on backbones trained with the default VQC head and 74.5 ± 0.5 on backbones trained with a linear head, a paired difference of -4.7 ± 0.9 points that is negative for all five seeds. The probe reads the VQC-trained bottleneck better than the VQC head itself does (67.8 ± 1.3 test): replacing the VQC head by the probe raises accuracy by about 2.0 points, and a 4.7-point gap remains between the two backbones under the common probe. The gap is larger with less data (CIFAR-10: -11.2 ± 4.0 and -11.1 ± 1.5 at 1,000 and 2,000 images) and smaller on MNIST (-7.9 ± 4.2, -2.2 ± 0.8 and -1.0 ± 0.5 at 250, 500 and 1,000; 3 seeds each). A narrow classical head lowers probe accuracy as well: MLP-8 backbones probe 3.6 ± 0.6 points below linear-head backbones on CIFAR-10/4,000, 1.0 ± 0.9 above the

Table 1. Test accuracy (%) of three heads on the same CNN, each with its head learning rate selected by mean validation accuracy over the tested grid (backbone lr is coupled at $0.5\times$ the head lr); mean $\pm$ SD over 3 seeds; bold: highest mean in the row. MLP-8: one hidden layer of width 8; VQC: the default 8-qubit, 4-layer ring circuit. Differences are in percentage points, paired over seeds (same training subset and split). The last column repeats VQC $-$ MLP-8 with both heads at one shared head lr (10^{-3}). The linear head's selected lr lies at the top of its grid in five rows, so its lead may be understated.

		Selected lr			VQC $-$ (selected lr)		VQC $-$ MLP-8
Data	N_{train}	Linear	MLP-8	VQC	Linear	MLP-8	(lr 10^{-3})
MNIST	250	$\mathbf{90.3 \pm 1.9}$	78.7 ± 6.5	76.6 ± 5.3	-13.7 ± 5.9	-2.1 ± 9.7	$+10.8 \pm 13.9$
	500	$\mathbf{94.3 \pm 0.2}$	86.7 ± 3.0	90.0 ± 0.6	-4.3 ± 0.5	$+3.3 \pm 2.6$	$+6.1 \pm 11.6$
	1000	$\mathbf{96.4 \pm 0.5}$	93.9 ± 0.8	94.6 ± 0.4	-1.7 ± 0.9	$+0.8 \pm 1.0$	-0.5 ± 3.8
CIFAR-10	1000	$\mathbf{59.0 \pm 0.9}$	52.3 ± 1.8	43.1 ± 3.8	-15.8 ± 3.8	-9.2 ± 4.5	-13.6 ± 1.8
	2000	$\mathbf{67.6 \pm 0.7}$	62.0 ± 1.0	53.1 ± 0.9	-14.5 ± 1.3	-8.9 ± 1.7	-13.5 ± 2.0
	4000	$\mathbf{74.6 \pm 0.5}$	70.8 ± 0.5	67.7 ± 1.1	-7.0 ± 1.2	-3.1 ± 0.9	-10.1 ± 1.6

VQC. The gap is not an artifact of linear decoding: a two-layer MLP probe on the same bottlenecks recovers at most 0.4 points more than the linear probe for any head, and puts the default VQC 4.8 ± 0.9 points below the linear head (Section D). As far as a linear and a two-layer probe can tell, the bottleneck's class information is linearly accessible, and the VQC-trained one carries less of it. The rest of this section shows that the VQC's share of the gap depends on its CNOT structure, which no classical head has.

5.2. Removing the CNOT rings recovers most of the gap

Deleting every CNOT and keeping the rest of the head unchanged raises probe accuracy to 73.6 ± 0.5 and test accuracy to 73.1 ± 0.5 (5 seeds). Against the default VQC this is $+3.8 \pm 1.2$ probe and $+5.2 \pm 1.4$ test points, positive for every seed, and the ring-free head now probes 2.8 ± 0.6 points above MLP-8. It leaves 0.8 ± 0.5 probe and 1.4 ± 0.5 test points to the linear head. MNIST shows the same direction: ring removal gains 2.8 ± 0.4 and 1.4 ± 0.4 probe points at 500 and 1,000 images, and the ring-free head's test accuracy (94.5, 96.4) equals the linear head's (94.3, 96.4) to within 0.2 points.

5.3. Product order tracks the gap

Which property of the rings matters? Figure 1b places every head at the product order of Equation (1). Across these eleven heads at four orders, the mean probe accuracy of every head is below that of every head of lower order. A ring after the last layer only keeps the default's parameter count and number of rotation layers but raises the order to 5.25; it lowers probe accuracy by 9.7 ± 2.2 points relative to the default (test -12.5 ± 2.3; 5 seeds), and the 1- and 2-layer circuits at the same order probe at 56.8 and 61.2 (3 seeds). Across these circuits, depth and parameter count do not order the cost; product order does.

Two interventions test the products directly. First, classical heads trained end to end on a circuit's zero-rotation products, with no circuit, land at the circuit's order on the trend: the products of the default circuit give 68.8 ± 1.6 probe and 67.1 ± 1.3 test, paired differences to the default VQC of -1.0 ± 2.2 and -0.7 ± 1.7 with mixed signs across seeds. Second, the *pairs* circuit keeps a CNOT after every layer but has order 1, because its pair CNOTs cancel at zero rotations. It recovers 3.1 ± 1.2 probe and 4.3 ± 1.2 test points over the default, positive for every seed, and sits 0.7 ± 1.0 probe points below the ring-free circuit. Pairs after the last layer only (order 1.5) probe at 71.2 ± 0.7, between orders 1 and 2.5, as does the classical head on their products (71.2 ± 1.3).

In CNOT-derived supports, order rises together with how many inputs different readouts share, so we also trained classical product heads on designed supports that separate the two (Table 7). Overlap, the mean number of inputs two readouts share, is equal for a "star" layout of order 3 and two order-4 layouts, and nearly equal for a star of order 2 and a window of three consecutive inputs (0.82 vs 0.86). In each of these comparisons the lower-order head probes higher: by 5.4 ± 2.7 points against an order-4 layout whose support matrix, like every CNOT-derived one, has full rank, 5.6 ± 2.2 against a rank-5 window, and 3.9 ± 1.9 for the order-2 star against the order-3 window (5 seeds, all positive). Order therefore matters at fixed overlap and full rank. At equal order, the stars, which put most inputs in a single product, probe 3.0 ± 0.2 (3 seeds) and 1.8 ± 2.8 (5 seeds, mixed signs) points above the windows; the first of these also differs in rank, so how the coupling is distributed across inputs matters in addition to order, by an amount these controls do not pin down. Order alone therefore does not set the cost: an order-3 star probes above the default ring's order-2.5 products (68.9 vs 67.8), and the designed heads also differ in how well they fit the training set (clean training accuracy 84.8% for the order-3 star vs 73.6% for the full-rank order-4 layout, seeds 0–2), so their contrasts mix structure with optimization.

The pattern also replicates on MNIST with 1,000 images, where we fixed the layouts and wrote down the predicted

ordering before running them. Probe accuracy is 96.5 without CNOTs, 95.5 for pairs after the last layer, 95.1 for the default circuit and 91.5 for a ring after the last layer; the small pairs-last advantage over the default is $+0.4 \pm 0.5$ (5 seeds, 4 positive), and the classical head on the default products probes 0.1 ± 0.5 below the default VQC, and the one on the last-ring products falls below that circuit (90.4).

The identity is exact only at zero rotations, so we check how far the trained circuits move from it (Section B), fitting an affine map from each circuit's zero-rotation readouts to its trained readouts on half of the validation set and evaluating it on the other half. A circuit frozen at its initialization is reproduced almost exactly ($R^2 = 0.997$). For the trained default and last-ring circuits the map recovers 87% and 90% of the held-out readout variance, and the backbone's cross-entropy gradient through the remapped products aligns with the trained one at median cosine 0.86 and 0.87. The products do not carry all task-relevant detail: the trained readout applied to the remapped products scores 7 and 6 points below the trained circuit, and without the remapping the gradient alignment is weaker (median 0.59 for the default circuit, and negative for one seed). Low-order circuits move further (R^2 0.73–0.83); for the ring-free circuit this is the $\sin a_j$ term of Equation (2), which its trained rotations use. At order 5.25 the trained rotations recover part of the cost: the last-ring circuit beats the classical head on its products by 6.4 ± 5.8 test points. The products thus reproduce the accuracy trend and most of the trained ring circuits' readouts and gradient directions, and the trained rotations add task detail on top of them.

5.4. Without CNOTs the head is a classical sinusoidal model

By Equation (2), the ring-free head computes $W[\alpha_j \cos a_j + \beta_j \sin a_j]_j + b$ for every setting of its rotations. Table 2 compares it with classical heads of this form trained under the same protocol. The cos-only restriction trails it by 0.3 ± 0.6 probe and 0.6 ± 0.7 test points with mixed signs, and the unrestricted cos+sin extension by 0.9 and 1.3 (3 seeds). The match holds at the level of the function class, not the parameters: the trained ring-free circuit has mean absolute R_y angle 0.41 rad and uses the sine component (Jacobian cosine 0.41 against the cos-only model; Table 6).

5.5. Controls that do not recover the gap

Table 3 tests alternative explanations on CIFAR-10/4,000. Lowering or raising the backbone learning rate with the head learning rate fixed leaves a gap of 4.9–8.6 points at every rate where the VQC trains, and the VQC collapses at the two highest rates. The backbone's gradient norm is larger, not smaller, with the VQC head. Freezing the circuit's rotations at initialization, which keeps the head close to its zero-rotation model, does not recover the probe (-1.6 ± 0.7 against the trained circuit). BatchNorm before the encoding and two alternative encodings, one of which breaks the sign symmetry of $\cos(\pi \tanh z)$, recover at most 0.7 probe points. Lowering product order gives by far the largest recoveries.

Higher-order circuits also fit their training data less well under the 80-epoch budget (clean training accuracy 96.6% without CNOTs, 89.2% for the default circuit and 65.6% for a ring after the last layer), so part of the gap could be under-training. Doubling the budget to 160 epochs for the linear, default, ring-free and last-ring heads (CIFAR-10/4,000) raises every probe but keeps the ordering (Table 8). The default circuit then fits its training set nearly as well as the ring-free and linear heads (98.8% vs 99.9%) and still probes 2.8 ± 0.4 points below the ring-free head and 3.1 ± 0.3 below the linear one (5 seeds, every seed negative). The last-ring circuit remains under-fit at 160 epochs (79.5%), so its larger cost includes slower optimization.

5.6. A pretrained backbone and the effect of data size

Hybrid image classifiers are often built on a pretrained network rather than a small CNN trained from scratch. We repeat the key heads on an ImageNet-pretrained ResNet-18 fine-tuned through each head, with its classifier replaced by the 8-dimensional bottleneck (Table 4). The ordering carries over unchanged. At 4,000 images the ring-free and pairs circuits probe within 0.2 points of the linear head, while the default circuit sits 7.2 ± 2.3 points below it and the last-ring circuit 18.0 ± 4.7 below (5 seeds, every seed negative); on the three seeds probed with both, a two-layer MLP probe gives the same gaps to within a point (default circuit: -5.8 vs -5.7). The cost depends strongly on data: the default circuit trails by 22.3 ± 2.9 points at 1,000 images and by 1.9 ± 0.4 at 10,000, while the ring-free head stays within 1.6 points of linear at every size. Small quantum heads are usually proposed for exactly the low-data regime where this cost is largest. One result differs from the small CNN: the classical head on the default circuit's products costs less than the circuit itself (-4.0 ± 1.6 vs -7.2 ± 2.3; the products probe 3.2 ± 1.3 points above the VQC, every seed positive), so on this backbone the products reproduce the ordering but only part of the circuit's cost.

6. Discussion and Limitations

A mechanism hypothesis. Why should product order cost the backbone? Differentiating Equation (1) at zero rotations gives, for $j \in S_k$,

$$\frac{\partial \langle Z_k \rangle}{\partial z_j} = -\pi \operatorname{sech}^2(z_j) \, \sin a_j \prod_{i \in S_k \setminus \{j\}} \cos a_i. \quad (3)$$

Table 2. Without CNOT rings the VQC head is exactly a classical per-coordinate sinusoidal model, $\text{logits} = W[\alpha_j \cos a_j + \beta_j \sin a_j]_{j=1}^{8} + b$ with $a_j = \pi \tanh z_j$ and α_j, β_j set by the trained rotations, for all rotation angles. Classical heads $W[\cos a_j]_j + b$ (a restriction) and $W[\cos a_j, \sin a_j]_j + b$ (an unrestricted extension), trained end to end with the same recipe and per-head validation lr selection, trail it by 0.3–0.9 probe points and 0.6–1.3 test points. CIFAR-10 with 4,000 training images; test accuracy of the full model and of the common linear probe on the bottleneck (%, mean ± SD over n seeds); differences in percentage points, paired over seeds.

				− ring-free VQC	
Head	n	Test	Probe	Test	Probe
VQC, no CNOT rings	5	73.1 ± 0.5	73.6 ± 0.5	—	—
cos-only (restriction)	5	72.5 ± 0.5	73.3 ± 0.4	-0.6 ± 0.7	-0.3 ± 0.6
cos+sin (unrestricted extension)[a]	3	72.0 ± 0.2	72.8 ± 0.4	-1.3 ± 0.4	-0.9 ± 0.5
linear (reference)	5	74.5 ± 0.7	74.5 ± 0.5	$+1.4 \pm 0.5$	$+0.8 \pm 0.5$

[a] 3 seeds; differences are against the ring-free VQC on the same seeds (test 73.3, probe 73.7).

The gradient a bottleneck coordinate receives through readout k is scaled by the cosines of every other coordinate in S_k, and it approaches zero when any of them nears $a_i = \pm\pi/2$. With order 1 each coordinate learns from its own readout; at order 5.25 the readouts couple 5.25 coordinates on average, multiplicatively. We hypothesize that this coupling makes the class signal each coordinate receives depend on the state of the others, so that higher-order heads train backbones whose bottlenecks carry less class signal that our probes can recover. Two results constrain the account. The MLP probe argues against the alternative that the information is present but only nonlinearly readable (Section D), consistent with the high-order circuits' weaker training fit. An attempt to remove the coupling directly, by keeping the forward products and passing each input's gradient only through its own factor, was inconclusive: the decoupled heads fit the training set far worse than the coupled ones (55.7% vs 84.0% clean training accuracy for the default supports), so the mismatched gradient, not the absence of coupling, dominates their result (Table 10). A test that preserves the true gradient, for example a reparameterization that changes the coupling while keeping the head's function, would isolate the mechanism.

Implications for hybrid design. For CNN-fed quantum heads trained end to end, the CNOT layout is a choice about the backbone: the zero-rotation supports, computable before training, order the mean probe accuracy of every circuit we test on both datasets, and of the three circuits we also train at twice the budget. In our setting the low-order heads are the best quantum heads, and the best of them, the ring-free head, is exactly a classical model by Equation (2). Head comparisons in this setting should tune each head separately, as Section 4 shows, and should measure the backbone, not only the head's accuracy.

Limitations. We study one circuit family (8 qubits, $R_y R_z$ layers, $\pi \tanh$ angle encoding with two variants), in noiseless simulation, with small CNNs and one pretrained ResNet-18 on subsets of MNIST and CIFAR-10 of at most 10,000 images; the size of the effect with other encodings, other pretrained models or on hardware is untested. Results use 3 seeds per cell and 5 for the CIFAR-10/4,000 headline cells, enough for consistent signs in paired comparisons but not for equivalence claims. Product order is a single scalar: it orders the cost at fixed overlap and full rank, but at fixed order the distribution of coupling across inputs also matters by an amount we do not pin down (Table 7), and the designed supports were tested with classical heads only. The identity is exact only at zero rotations; the trained default and last-ring circuits sit at $R^2 = 0.87$–0.90 of it, and low-order circuits move further. Finally, our probes are linear and a two-layer MLP; information recoverable only by stronger decoders is not measured.

7. Conclusion

A quantum classifier head trained jointly with a CNN changes the representation the CNN learns. With every head tuned on validation data and every backbone read by the same linear probe, the default ring-entangled VQC head lowers CIFAR-10 probe accuracy by 4.7 points, and the drop follows a quantity fixed by the CNOT pattern: the order of the cosine products each readout computes at zero rotations. Lowering that order, by removing the rings or by entangling only in pairs that cancel at zero rotations, recovers most of the gap, and classical heads built from the same products reproduce the declining probe-accuracy trend. The pattern holds on a pretrained ResNet-18, where the cost of the default head is largest with the least data. Without CNOTs the quantum head is a classical sinusoidal model. For practitioners, three steps follow: tune each head separately, probe the backbone, and count the product order of a circuit before training through it.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the evaluation of hybrid quantum–classical classifiers. Its practical consequence is

Table 3. Controls on CIFAR-10 with 4,000 training images. Entries are differences in the test accuracy of the common linear probe on the frozen 8-dimensional bottleneck, in percentage points, paired over 3 seeds (mean ± SD). The five controls leave a substantial probe gap; removing the CNOT rings recovers most of it.

Control	Comparison	Probe Δ	Observation
Reference: default VQC	VQC − linear	-4.7 ± 1.2	the gap under study
Backbone lr[a]	VQC − linear, backbone lr 0.0025	-6.3 ± 1.2	gap persists at every tested backbone lr where the VQC trains
	VQC − linear, backbone lr 0.005	-4.9 ± 1.2	
	VQC − linear, backbone lr 0.015	-8.6 ± 0.5	
Freeze circuit at init	frozen − trained circuit	-1.6 ± 0.7	gap persists with rotations frozen at initialization
BatchNorm before encoding	VQC+BN − VQC	-1.2 ± 2.2	pre-encoding BN does not recover the probe
	VQC+BN − linear	-5.9 ± 3.4	
Monotone encoding[b]	monotone − default	$+0.3 \pm 0.7$	neither encoding recovers more than 0.7 points
	symmetric control − default	$+0.7 \pm 0.5$	
	monotone − symmetric control	-0.4 ± 0.9	
Contrast: remove CNOT rings	ring-free − default	$+3.8 \pm 1.6$	recovers most of the gap (§5.2)
	ring-free − linear	-0.8 ± 0.7	

[a]Head lr fixed at 0.01 (VQC) and 0.03 (linear) while the backbone lr varies; the VQC collapses at the tested backbone lrs 0.03 and 0.06. Other rows: each head at its validation-selected head lr, backbone lr $0.5\times$ head lr. [b]Monotone: $R_y(\frac{\pi}{2}\tanh z + \frac{\pi}{2})$; symmetric control: $R_y(\frac{\pi}{2}\tanh z)$, same range width; default: $R_y(\pi \tanh z)$. Not a probe difference: the final-epoch pre-clip backbone gradient norm is larger with the VQC head than with the linear head (1.92 ± 0.13 vs 0.75 ± 0.02).

methodological: comparisons between quantum and classical heads should tune each head separately and measure the representation the head induces. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Table 4. The representation cost on an ImageNet-pretrained ResNet-18 fine-tuned through each head (CIFAR-10; backbone learning rate $0.1\times$ the head's; 30 epochs at 128 px). Probe: test accuracy (%) of the common linear probe on the frozen 8-dimensional bottleneck, and its paired difference to the linear head in percentage points (mean $\pm$ SD; 4,000 images: number of seeds in parentheses; 1,000 and 10,000 images: 3 seeds). Head learning rates are selected on validation separately at each size. At 4,000 images every head's grid spans 3×10^{-4}–10^{-2}, extended to 10^{-4} for the default and last-ring circuits; at 10,000 images it spans 3×10^{-4}–3×10^{-3}, extended to 10^{-4} for the linear, default and last-ring heads. At both sizes every selected rate is interior. The last column repeats the 4,000-image difference with a two-layer MLP probe (3 seeds).

		1,000 images	4,000 images		10,000 images	4,000, MLP probe
Head	Order	− linear	probe	− linear	− linear	− linear
linear	—	82.9 ± 0.9	89.4 ± 0.4 (5)	—	92.6 ± 0.4	—
VQC, no CNOT	1	-1.6 ± 1.2	89.4 ± 0.4 (5)	0.0 ± 0.1	-0.1 ± 0.4	-0.6 ± 0.3
VQC, pairs	1		89.3 ± 0.4 (5)	-0.2 ± 0.3		-0.8 ± 0.5
cos-prod. (default supports)	2.5		85.5 ± 1.6 (5)	-4.0 ± 1.6		-3.0 ± 1.1
VQC, ring (default)	2.5	-22.3 ± 2.9	82.3 ± 2.3 (5)	-7.2 ± 2.3	-1.9 ± 0.4	-5.8 ± 0.3
VQC, ring, last	5.25	-34.1 ± 6.3	71.4 ± 4.8 (5)	-18.0 ± 4.7	-8.0 ± 1.0	-18.5 ± 3.9

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A. The Zero-Rotation Identity

Let $U(\theta)$ be the variational block and $|\psi(\mathbf{z})\rangle = \bigotimes_j R_y(a_j)\,|0\rangle$ the encoded product state, so that $\langle Z_k\rangle = \langle\psi|\,U^\dagger Z_k U\,|\psi\rangle$. At $\theta = 0$ every R_y and R_z is the identity and U is the product of the circuit's CNOTs. Conjugation by $\mathrm{CNOT}(c,t)$ maps $Z_t \mapsto Z_c Z_t$ and fixes Z_c, so $U^\dagger Z_k U$ is obtained by applying these rules to Z_k for the gates in reverse order; the result is a string $\prod_{j\in S_k} Z_j$, with S_k the set of qubits on which an odd number of propagation paths land. Since $|\psi\rangle$ is a product state with $\langle\psi|\,Z_j\,|\psi\rangle = \cos a_j$, the expectation factorizes into Equation (1). Table 5 lists the supports. Pairs after every layer apply each pair CNOT four times, which is the identity, so their supports equal the no-CNOT ones; with trained rotations between the layers the gates no longer cancel, and the circuit produces entangled states (we checked numerically that its outputs differ from the ring-free circuit's at random rotation angles).

For the ring-free circuit and any rotation angles, qubit j starts from Bloch vector $(\sin a_j, 0, \cos a_j)$ after encoding and then undergoes a fixed rotation $M_j \in \mathrm{SO}(3)$ determined by its trained $R_y R_z$ layers. Hence $\langle Z_j\rangle = (M_j)_{zx}\sin a_j + (M_j)_{zz}\cos a_j$, which is Equation (2) with $\beta_j = (M_j)_{zx}$ and $\alpha_j = (M_j)_{zz}$. We verified the formula against the simulator to 10^{-7}. Since the readout layer is linear, the ring-free head is $W[\alpha_j \cos a_j + \beta_j \sin a_j]_j + b$.

Table 5. Zero-rotation supports S_k of each readout $\langle Z_k\rangle$, written as digit strings (e.g. 015 is $\{0,1,5\}$), and the mean product order $\bar{s}$, computed by propagating Z_k backwards through each circuit's CNOTs.

Circuit	$\bar{s}$	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
no CNOT	1	0	1	2	3	4	5	6	7
pairs (every layer)	1	0	1	2	3	4	5	6	7
pairs, last	1.5	0	01	2	23	4	45	6	67
ring (default)	2.5	015	256	367	047	15	26	37	04
ring, last	5.25	1234567	01	012	0123	01234	012345	0123456	01234567
ring, $L = 1$	5.25	1234567	01	012	0123	01234	012345	0123456	01234567
ring, $L = 2$	5.25	01357	0234567	134567	024567	13567	02467	1357	0246

B. Trained Circuits against the Product Model

Table 6. Trained circuits against their own zero-rotation product model (CIFAR-10, 4,000 training images; each run's validation embeddings, split by index parity; means over n seeds). An affine map from the readouts of the same circuit with all rotations set to zero, q_0, to the trained readouts q is fit on one half and evaluated on the other. R^2: held-out fraction of trained readout variance it recovers, a ratio relative to the circuit's own output spread; MSE: the unnormalized held-out mean squared error per readout ($\langle Z\rangle \in [-1,1]$). Accuracy: held-out accuracy of the trained head, and of its trained readout layer applied to the remapped q_0. Loss-gradient cosine: median over held-out samples of the cosine between $\partial\mathcal{L}/\partial z$ (cross-entropy) of the trained head and of the zeroed head with the trained readout kept (zeroed) or applied to the remapped q_0 (remapped). Last column: mean $|\theta|$ of the R_y / R_z rotations (rad).

					Accuracy		Loss-grad. cosine		
Circuit	Order	n	R^2	MSE	trained	remapped	zeroed	remapped	$\lvert R_y\rvert$ / $\lvert R_z\rvert$
ring (default), frozen at init	2.5	3	0.997	0.0010	67.6	67.0	1.00	1.00	0.05 / 0.05
ring (default)	2.5	5	0.866	0.0414	69.3	62.3	0.59	0.86	0.14 / 0.48
ring, last	5.25	5	0.903	0.0326	55.7	49.4	0.75	0.87	0.10 / 0.41
pairs, last	1.5	5	0.826	0.0684	70.0	60.4	0.49	0.81	0.20 / 0.37
pairs	1	5	0.788	0.0990	73.9	64.3	0.42	0.79	0.32 / 0.43
no CNOT	1	5	0.734	0.1309	74.1	61.9	0.34	0.71	0.41 / 0.28

Table 6 compares each trained circuit with the same circuit after setting every rotation to zero, on the validation embeddings of the run's own backbone. The frozen circuit anchors the measurement: rotations held at their $\mathcal{U}(-0.1, 0.1)$ initialization reproduce the product model almost exactly. Held-out R^2 equals the in-sample value to within 0.015 for every run, so the affine map does not overfit. These are endpoint measurements: they say how close the trained circuit is to its product model, not that training followed the product model's trajectory. Two results bound the approximation. First, the products, even after remapping, lose task-relevant detail: held-out accuracy drops by 6–12 points. Second, gradient alignment depends on the remapping: with the trained readout applied directly to the zeroed circuit, the median loss-gradient cosine is 0.59 for the default circuit, and -0.08 for its seed 4 (0.96 after remapping); after remapping it is 0.79–0.87 for every entangling circuit, with 3–8% of samples negative on average (18% in the worst run). Zeroing the rotations without remapping costs 14–23 accuracy points on average for the two ring circuits and 31–43 for the pairs and CNOT-free circuits, which mostly reflects

the unrefitted readout. Low-order circuits drift further, as Section 5.4 explains for the ring-free case.

C. Order versus Overlap, and Training Budget

Table 7. Classical product heads whose supports separate product order from overlap (mean pairwise $|S_k \cap S_l|$), trained end to end like every other head (CIFAR-10, 4,000 training images; n seeds, each at its validation-selected learning rate; differences paired over shared seeds). Rank: rank of the 8×8 support matrix, which bounds the number of independent input directions the products can respond to. Window w: $S_k = \{k, \ldots, k{+}w{-}1\}$; gapped window 4: $\{k, k{+}1, k{+}2, k{+}4\}$ (mod 8). Star 2: $\{1,2\}, \{0,1\}, \{0,2\}, \ldots, \{0,7\}$. Star 3: $\{1,2,3\}, \{0,1,2\}, \{0,1,3\}, \ldots, \{0,1,7\}, \{0,2,3\}$. Paired probe differences: star 3 − gapped window 4 (equal overlap, both rank 8): $+5.4 \pm 2.7$; star 2 − window 3 (approximately equal overlap, both rank 8): $+3.9 \pm 1.9$; star 3 − window 4 (equal overlap, rank 8 vs 5): $+5.6 \pm 2.2$; star 2 − window 2 (equal order, rank 8 vs 7): $+3.0 \pm 0.2$; star 3 − window 3 (equal order, both rank 8): $+1.8 \pm 2.8$.

Supports	Order	Overlap	Rank	n	Test	Probe
cos (order 1)	1	0.00	8	3	72.2 ± 0.3	73.2 ± 0.1
window 2	2	0.29	7	3	68.1 ± 0.4	69.3 ± 0.4
star 2	2	0.82	8	5	69.8 ± 2.2	70.9 ± 1.9
default ring products	2.5	0.57	8	3	66.3 ± 0.9	67.8 ± 1.1
window 3	3	0.86	8	5	64.2 ± 2.9	67.1 ± 2.3
star 3	3	1.71	8	5	67.2 ± 1.2	68.9 ± 1.1
window 4	4	1.71	5	5	60.1 ± 3.9	63.3 ± 3.1
gapped window 4	4	1.71	8	5	61.0 ± 2.4	63.5 ± 1.8

Table 8. Twice the training budget (160 instead of 80 epochs, each head at its 80-epoch learning rate and cosine schedule; CIFAR-10, 4,000 training images; n seeds). Clean-train: accuracy on the unaugmented training subset at the selected checkpoint; loss: final-epoch training loss on augmented batches. At 160 epochs, default − no-CNOT probe accuracy is -2.8 ± 0.4 and default − linear is -3.1 ± 0.3 (paired).

Head	Epochs	n	Clean-train	Train loss	Test	Probe
linear	80	5	98.7 ± 0.4	0.13	74.5 ± 0.7	74.5 ± 0.5
	160	5	99.9 ± 0.1	0.03	75.6 ± 0.6	75.5 ± 0.4
no CNOT	80	5	96.6 ± 0.7	0.22	73.1 ± 0.5	73.6 ± 0.5
	160	5	99.9 ± 0.1	0.03	74.6 ± 0.5	75.1 ± 0.4
ring (default)	80	5	89.2 ± 2.5	0.44	67.8 ± 1.3	69.8 ± 1.0
	160	5	98.8 ± 0.6	0.08	70.6 ± 0.5	72.4 ± 0.6
ring, last	80	5	65.6 ± 5.4	1.02	55.3 ± 3.1	60.1 ± 3.1
	160	3	79.5 ± 15.1	0.67	59.6 ± 7.6	64.7 ± 4.7

Table 7 lists the designed-support heads of Section 5.3. Overlap is the mean over ordered readout pairs $k \neq l$ of $|S_k \cap S_l|$; if input j appears in d_j supports, order is $\frac{1}{8}\sum_j d_j$ and overlap is $\frac{1}{56}\sum_j d_j(d_j - 1)$, so for windows, where every d_j equals the order, the two rise together. Stars hold the order while concentrating the d_j on one or two inputs. Windows of width 2 and 4 have rank-deficient support matrices; the gapped window of width 4 restores full rank at the same order and overlap and probes like the rank-5 window (-0.7 on average, mixed in sign), so rank does not account for the order-4 deficit. Table 8 gives the training-budget check of Section 5.5, with the linear head also rerun at 160 epochs.

D. Nonlinear Probe and Gradient Decoupling

Table 9 refits every headline backbone of Figure 1b with a two-layer MLP probe under the same validation protocol as the linear probe; MNIST with 1,000 images gives the same picture (the MLP probe is within 0.5 points of the linear probe for every head). Table 10 gives the decoupling attempt discussed in Section 6.

E. Exploratory Sweep with One Shared Recipe

Before the protocol of Section 3, we ran every head with one shared recipe: head learning rate 10^{-3} (backbone 5×10^{-4}), 25 epochs on MNIST and 80 on CIFAR-10, three data re-uploading schedules for the VQC, and checkpoints chosen by best test-split accuracy. Because the test split took part in checkpoint selection, these numbers are exploratory; mean final-epoch accuracy is within 1.35 points of best-epoch accuracy in all 30 configurations and no ordering of heads changes (`scripts/make_tables.py` on the released per-epoch logs in `results/raw`). Tables 11 and 12 give the results.

Table 9. Linear vs nonlinear probe on the same frozen bottlenecks (CIFAR-10, 4,000 training images; MLP probe: standardization, then $8 \to 256 \to 256 \to 10$ with ReLU, probe learning rate and epoch selected on validation; mean $\pm$ SD over n seeds; last column paired). The MLP probe recovers at most a fraction of a point more than the linear probe for any head, and VQC $-$ linear is -4.8 ± 0.9 under the MLP probe vs -4.7 ± 0.9 under the linear probe.

Head the backbone was trained with	n	Linear probe	MLP probe	MLP $-$ linear
linear	5	74.5 ± 0.5	74.4 ± 0.5	0.0 ± 0.1
MLP-8	5	70.8 ± 0.5	70.8 ± 0.6	0.0 ± 0.6
VQC, no CNOT	5	73.6 ± 0.5	73.6 ± 0.4	0.0 ± 0.4
VQC, pairs	5	72.9 ± 0.6	72.9 ± 0.5	0.0 ± 0.4
VQC, pairs, last	5	71.2 ± 0.7	71.2 ± 0.5	0.0 ± 0.4
VQC, ring (default)	5	69.8 ± 1.0	69.6 ± 0.8	-0.2 ± 0.3
VQC, ring, last	5	60.1 ± 3.1	60.3 ± 2.7	$+0.2 \pm 0.9$
cos	5	73.3 ± 0.4	73.2 ± 0.6	-0.2 ± 0.3
cos-prod. (default supports)	5	68.8 ± 1.6	69.0 ± 1.5	$+0.2 \pm 0.2$
cos-prod. (last-ring supports)	5	54.8 ± 3.4	55.2 ± 3.3	$+0.4 \pm 0.4$

Table 10. Gradient-decoupled classical product heads: same forward products, but each input's gradient passes only through its own cosine factor (straight-through on the other factors). CIFAR-10, 4,000 images, 3 seeds; decoupled heads at their validation-selected learning rate (grid 10^{-3}–3×10^{-2}), coupled heads as in Table 7. Decoupling fails to fit the training set, so it does not isolate the effect of coupling.

		Clean-train acc.		Probe	
Products	lr	coupled	decoupled	coupled	decoupled
default supports	0.003	84.0 ± 1.7	55.7 ± 3.7	67.8 ± 1.1	57.7 ± 1.5
last-ring supports	0.003	56.4 ± 5.2	42.4 ± 5.3	54.2 ± 3.1	48.9 ± 4.7

Re-uploading lowered accuracy in every setting under this recipe (NONE $>$ LIGHT $>$ STRONG); it was not part of the tuned protocol.

Table 11. Exploratory sweep, MNIST: best test-split accuracy (%, mean $\pm$ SD over 3 seeds) with one shared recipe. Bold marks the better of MLP-8 and the VQC (NONE).

			VQC (re-uploading schedule)		
N_{train}	Linear	MLP-8	NONE	LIGHT	STRONG
250	61.5 ± 1.7	36.7 ± 5.4	$\mathbf{54.5 \pm 3.7}$	48.3 ± 10.7	38.3 ± 7.4
500	79.3 ± 2.9	53.2 ± 6.9	$\mathbf{71.2 \pm 1.9}$	57.8 ± 2.5	45.6 ± 6.7
1,000	89.6 ± 0.7	76.4 ± 3.4	$\mathbf{83.3 \pm 0.3}$	68.4 ± 3.5	54.6 ± 14.0

F. Frozen-Embedding Transfer

To separate a head from the backbone it induces, we also trained fresh heads on fixed embeddings extracted from exploratory-sweep backbones (linear- or VQC-trained; MNIST at 1,000 images, CIFAR-10 at 4,000; 3 seeds; 498 head fits with learning rate and epoch selected on validation). Table 13 gives the results. On VQC-trained embeddings, a refitted linear head beats a refitted VQC by 1.7 ± 0.8 (MNIST) and 1.6 ± 0.5 points (CIFAR-10), positive for every seed. On linear-trained embeddings, most of whose values lie in the flat region $|\tanh z| > 0.9$ of the encoding (56% and 66%), the VQC collapses, trailing the linear head by 20.9 ± 4.4 and 41.8 ± 2.9 points, and a classical Fourier head $[\cos a_j, \sin a_j] \to$ Linear with the same encoding collapses almost as far on CIFAR-10. Fixed standardization of the embedding recovers 12.6 ± 6.2 and 30.4 ± 2.9 of those points for the VQC, and a learned per-dimension affine 15.0 ± 7.6 and 35.9 ± 0.9; tripling the training budget recovers 0.5 ± 0.2 on CIFAR-10. The Fourier head's recovery under standardization was not tested. The source checkpoints were selected on the test split, so this experiment is exploratory; it motivated the probe measurements of Section 5, which use validation-selected backbones only.

Table 12. Exploratory sweep, CIFAR-10: best test-split accuracy (%, mean ± SD over 3 seeds) with one shared recipe. Bold marks the better of MLP-8 and the VQC (NONE).

			VQC (re-uploading schedule)		
N_{train}	Linear	MLP-8	NONE	LIGHT	STRONG
1,000	57.5 ± 0.8	$\mathbf{51.2 \pm 0.8}$	37.1 ± 2.4	34.2 ± 0.7	26.6 ± 1.9
2,000	64.6 ± 0.7	$\mathbf{60.6 \pm 0.9}$	46.8 ± 5.2	42.7 ± 3.9	28.7 ± 1.9
4,000	71.2 ± 0.5	$\mathbf{67.2 \pm 0.4}$	58.9 ± 4.6	52.0 ± 7.8	35.2 ± 5.5

Table 13. Frozen-embedding transfer: test accuracy (%, mean ± SD over 3 seeds) of heads trained on identical frozen embeddings. Columns give the head the backbone was trained with and the fraction of test embedding values with $|\tanh z| > 0.9$. †Selected learning rate is the largest tried for that head.

	MNIST (1,000 images)		CIFAR-10 (4,000 images)	
Head	linear src. (56%)	VQC src. (5%)	linear src. (66%)	VQC src. (30%)
Linear	89.8 ± 0.2	88.9 ± 1.6	70.9 ± 0.6	$62.5 \pm 4.4^{\dagger}$
MLP-8	88.3 ± 0.9	88.5 ± 1.1	70.2 ± 0.8	62.3 ± 4.0
Fourier	78.2 ± 1.1	87.1 ± 2.0	33.4 ± 1.0	61.8 ± 4.5
VQC	69.0 ± 4.4	87.2 ± 1.0	29.1 ± 3.0	61.0 ± 4.1
VQC+std	81.6 ± 5.6	77.6 ± 6.6	59.5 ± 0.4	53.7 ± 5.6
VQC+std+affine	84.0 ± 3.3	82.7 ± 1.6	65.0 ± 3.1	61.6 ± 4.4
VQC, 3× epochs	–	–	$29.6 \pm 3.0^{\dagger}$	$61.0 \pm 4.5^{\dagger}$

G. Learning-Rate Sensitivity

Trigonometric heads, quantum or classical, tolerate a narrow range of head learning rates under the coupled recipe (backbone at half the head rate). On CIFAR-10/4,000 all of them select 10^{-2}, and at 3×10^{-2} test accuracy falls to 63.4 for the default VQC, 50.0 for the ring-free VQC, and 29.3 and 28.9 for the classical cos and cos+sin heads, while the linear head selects 3×10^{-2}. With the head learning rate fixed, the default VQC also collapses at backbone learning rates of 3×10^{-2} and 6×10^{-2}, where its bottleneck saturates the encoding completely, whereas the linear head still trains (Table 3). Test accuracy for every grid point is in the released results.

H. Sign Symmetry

At zero rotations every readout is an even function of each z_j, because cosine is even. We measured how far trained heads are from this symmetry on CIFAR-10/4,000 test embeddings: the median, over samples, of $\|q(-\mathbf{z}) - q(\mathbf{z})\| / \|q(\mathbf{z}) - q(\mathbf{z}')\|$, with q the readouts and $\mathbf{z}'$ a random other sample (0: blind to the sign; about 1: flipping the sign moves the readout as far as changing the image). The frozen circuit stays nearly sign-blind (0.09–0.12), and its accuracy on $-\mathbf{z}$ is within 1.4 points of that on $\mathbf{z}$. Trained heads are not: 0.58–0.73 for the default VQC (accuracy on $-\mathbf{z}$: 25–34%), 0.91–1.02 for the ring-free VQC (6–17%), and 0.85–0.97 for the monotone and symmetric encodings of Table 3. Sign-blindness at initialization therefore does not predict the backbone cost, consistent with the null result of the monotone encoding.

I. Training and Reproducibility Details

Training. AdamW with weight decay 10^{-4}, cosine annealing to zero, batch size 32, gradient clipping at norm 1.0 (MNIST) or 2.0 (CIFAR-10); CIFAR-10 augmentation is random crop, horizontal flip, small rotation and color jitter. Head learning-rate grids: MNIST $\{3 \times 10^{-4}, \ldots, 3 \times 10^{-2}\}$ and CIFAR-10 $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$ in half-decade steps, each extended by one step whenever the selected rate was at the top; circuit variants $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$; classical product heads $\{10^{-3}, \ldots, 3 \times 10^{-2}\}$. All selected rates of circuit and classical product heads are interior to their grids. The probe uses AdamW (weight decay 10^{-4}), batch size 128, 60 epochs with a cosine schedule, and probe learning rates $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}, 3 \times 10^{-1}, 1\}$, all selected on validation.

Reproducibility. The repository contains per-epoch metrics and metadata for every run, the probe and trained-circuit measurements, and the scripts that regenerate every number, table and figure from them (`scripts/summarize_v2.py`, `scripts/zero_angle_check.py`, `paper/figures/scripts/`). The PyTorch statevector simulator is tested

against PennyLane to 10^{-6} in outputs and gradients. Seeds control data subsets, initialization and data order. Experiments ran on one Apple M3 Max.